

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0241

Barrett 500 kV
January 17, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0241. The Interconnection Customer proposes a 390.7 MW solar photovoltaic facility interconnecting at the Barrett 500 kV substation in the Ohio area. The request was received on May 29, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 163 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,144,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Fishkill Plains to Dillard-Orange 345 kV is overloaded to 118.9% of its 1,399 MVA emergency rating for the loss of the Flatlick to International Paper Substation 500 kV. The project contributes 6.1% DFAX (24.0 MW) to this overload.

The Winthrop to Minco 230 kV is overloaded to 115.5% of its 546 MVA emergency rating for the loss of the Barrett to Norwich 500 kV L. The project contributes 9.0% DFAX (35.2 MW) to this overload.

Hampton Rd to Medora 345 kV: 109.9% loading (1,243 MVA) following the outage of the Simpson Stuart Tap to Fishkill Plains 345 kV; DFAX = 7.8%.

The Medora to P.T.S 345 kV is overloaded to 107.9% of its 1,258 MVA emergency rating for the loss of the Ritter Park to Sugar Hill 345 kV Line. The project contributes 8.6% DFAX (33.4 MW) to this overload.

The Sugar Hill to Ritter Park 345 kV is overloaded to 104.5% of its 1,250 MVA emergency rating for the loss of the Hilltop to Piney Crk Project Substation 500 kV. The project contributes 10.0% DFAX (39.2 MW) to this overload.

For the loss of P.T.S to Medora 345 kV, the Stewart Mtn Dam to Ritter Park 345 kV loads to 104.4% of its 1,173 MVA rating. Project DFAX: 7.8%.

The Breeze to Barrett 230 kV is overloaded to 101.0% of its 621 MVA emergency rating for the loss of the Mossyrock 500 kV/230 kV Transformer. The project contributes 5.2% DFAX (20.5 MW) to this overload.

Burwell 500-345 kV XFMR: 100.8% loading (692 MVA) following the outage of the Allegheny Energy Units 3 & 4 & 5 345/1 kV TX; DFAX = 6.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fishkill Plains to Dillard-Orange 345 kV	Rebuild line with higher-capacity conductor	\$95,126,000	4% (\$4,312,000)
Winthrop to Minco 230 kV	Rebuild line with higher-capacity conductor	\$10,367,000	6% (\$610,000)
Hampton Rd to Medora 345 kV	Reconductor line	\$9,464,000	6% (\$563,000)
Medora to P.T.S 345 kV	Reconductor line	\$9,928,000	6% (\$591,000)
Sugar Hill to Ritter Park 345 kV	Reconductor line	\$8,887,000	6% (\$535,000)
Ritter Park to Stewart Mtn Dam Substation 345 kV Line	Reconductor line	\$7,114,000	6% (\$425,000)
Breeze to Barrett 230 kV	Reconductor line	\$5,265,000	8% (\$446,000)
Burwell 500/345 kV TX	Replace transformer with higher-rated unit	\$27,873,000	6% (\$1,662,000)
	Total allocated to project		\$9,144,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.